

Trainer Debrief — Lab 16

Fishbone, 5 Whys and Cause Prioritisation — In Depth

Case study: Meridian Medical Devices — seal weld burst pressure

Objective: Generate and prioritise candidate root causes for your capstone (A3, K2).

***TRAINER ONLY.** Do not hand this out before the lab is complete.*

Expected analysis

lab-16-cause-prioritisation.xlsx

Scored against the Y with a C&E matrix, the top causes are fixture clamp pressure, weld dwell time and electrode tip wear. These become the Xs you test in Labs 17-20 and manipulate in the DOE in Labs 23-25. Note that 'no written burst-test procedure' is a MEASUREMENT cause already confirmed by the failed Gage R&R in Lab 11 — fix it, but it is not a driver of the true Y.

Every dataset is generated from a fixed seed, so these figures are reproducible — re-run the analysis on the shipped workbook and you will get these numbers.

What good looks like

Your fishbone carries 25+ causes across all six bones, 5 Whys chains terminate at process rather than person, and you have five testable causes written as formal hypotheses.

Where this sits in the capstone

Re-derive cause-and-effect analysis in full — to the depth at which you can facilitate it for a Green Belt team, which is a core Black Belt accountability. The output is a prioritised, testable cause list, not an unmanageable wall of sticky notes. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

The running process — figures that must reconcile

Checkpoint	Value	Established in
Baseline defect rate	2.9%	Lab 3
Measurement system	FAILS Gage R&R (~40% study variation)	Lab 11
Baseline capability	Pp 0.62 / Ppk 0.53	Lab 14
Proven Xs	clamp pressure, dwell time, electrode tip age	Labs 18-21
DOE optimum	clamp 47.4 bar, dwell 2.76 s	Labs 23-25
Improved capability	Ppk 1.40	Lab 27
Annual hard benefit	SGD 487,000 vs SGD 812,000 COPQ	Labs 1, 30

If a learner's number disagrees with the column above, they have either used their own workplace data (fine — say so) or made an error (find it).

Certified Lean Six Sigma Black Belt (CLSSBB) Training · TGS-2024051900 · Version v4 · © 2026 Tertiary Infotech Academy Pte Ltd · TRAINER ONLY